

September dataset and experiment update

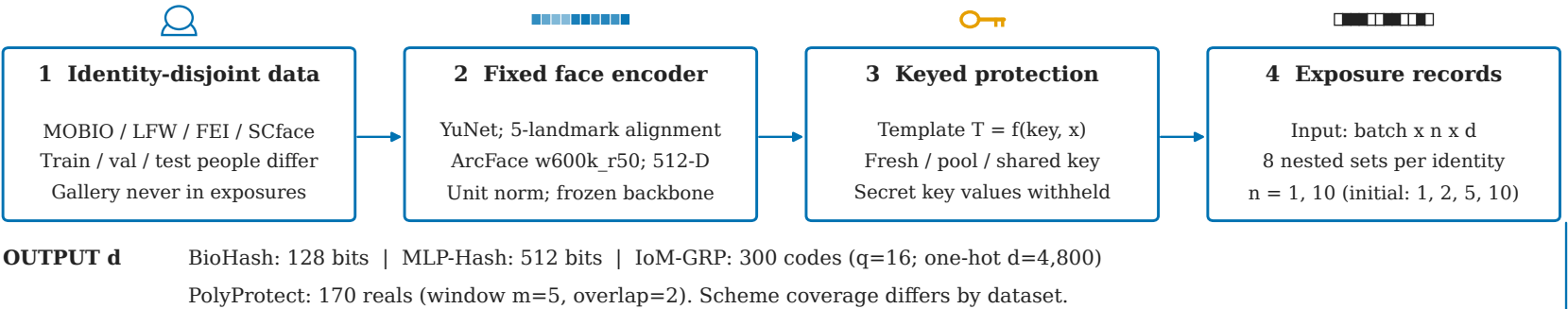
Identity-disjoint evaluation across session, pose and camera variation.

Dataset	Identities	Train / val / test	Valid / selected	Variation
MOBIO	150	90 / 30 / 30	1,799 / 1,800	Sessions
LFW	125	75 / 25 / 25	1,500 / 1,500	In-the-wild
FEI	200	120 / 40 / 40	2,378 / 2,400	Pose / expression
SCface	130	78 / 26 / 26	2,851 / 2,860	Camera / distance

SCface: mugshot-to-surveillance linkage under capture-domain shift.
Mugshot gallery / surveillance exposures; all 130 identities remain eligible after 9 detection failures.
Unprotected SCface top-1: 84.375% over 544 test probes; chance: $1/26 = 3.846\%$.
Evidence: 73 key-pool conditions / 12 studies; 72 one-seed model endpoints / 24 scheme pilot cells.
New MOBIO/FEI follow-up: 216 trained endpoints; 3 model seeds x 2 identity partitions.

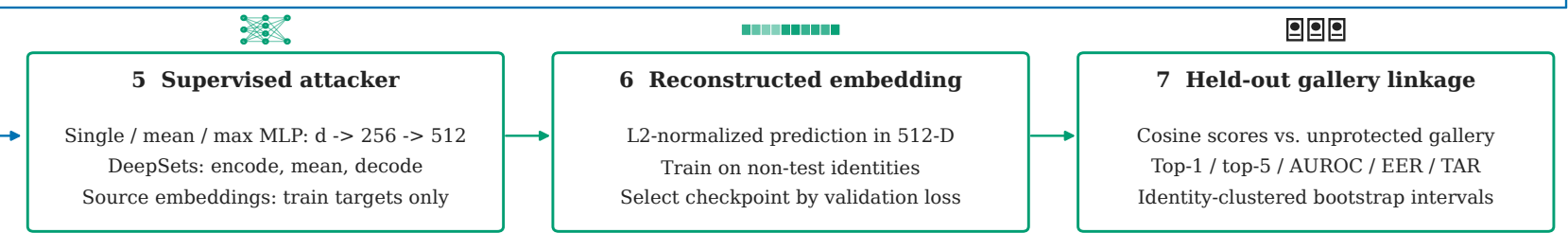
Experimental architecture

A DATA AND PROTECTION



B KEY-BLIND ATTACK AND EVALUATION

Attacker observes protected records, not keys or slot labels; slots are revealed only in a labelled control.



- KEY DESIGN** Fresh: source-record keys are split-disjoint. Pool K: hidden transforms recur across identity splits.
- REPLICATION** MOBIO / FEI follow-up: 2 identity partitions x 3 model seeds; matched 120-epoch training caps.
- DESIGN** Same-identity aggregation under controlled transform reuse; detailed attacker paths in the companion figure.

System overview: fixed encoder, hidden-key protection, same-identity exposure sets and held-out gallery linkage.

Detailed attacker: aggregation and reconstruction

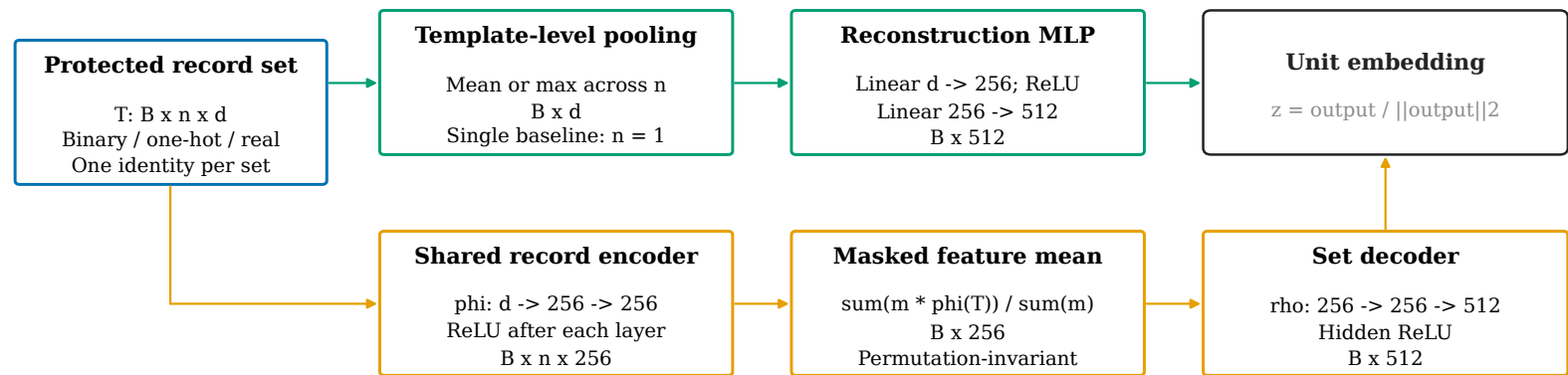
A MATCHED EXPOSURE CONSTRUCTION

One identity, n source records; nested subsets from one fixed permutation per repeat; gallery excluded.

Fresh: independent source-record keys. Pool K: hidden transforms recur across identity splits. No keys or slots in model input.

T: $B \times n \times d$ $d = 128$ BioHash; 512 MLP-Hash; 4,800 one-hot IoM-GRP; 170 PolyProtect

B TWO AGGREGATION PATHS



Primary: mean-pool MLP. Secondary: DeepSets. m masks valid records; all n records are valid in these runs.

C SUPERVISION

Target $u = \text{normalize}(\text{mean of exposed source embeddings})$.

Loss = $\text{mean}[1 - \cos(z, u)] + 0.1 \times \text{MSE}(z, u)$.

Adam: lr 0.001; weight decay 0.0001; hidden width 256.

Train identities only; select minimum validation loss; freeze test.

D HELD-OUT IDENTITY LINKAGE

Scores = $z^T G$; G : $N \times 512$ unprotected unit gallery.

Rank identities; top-1/top-5, AUROC, EER, TAR at FAR.

Follow-up: 3 model seeds x 2 identity partitions; 120 epochs.

Crossed seed/identity intervals; paired primary contrasts.

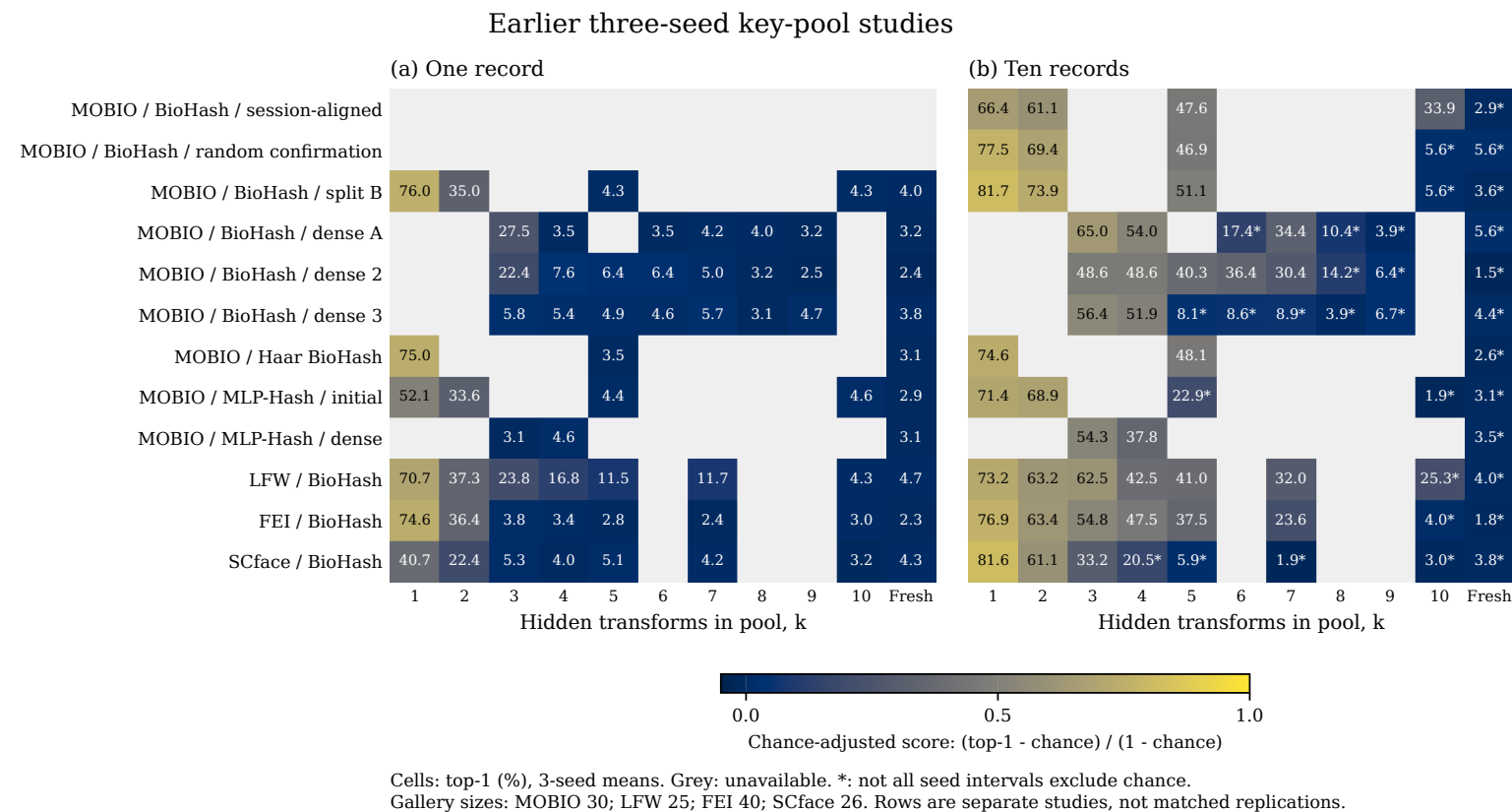
B: batch size; n: exposure count; d: template input width; N: test gallery identities. Source targets are not inference inputs.

Mean pooling is the primary endpoint; DeepSets is secondary. Target is the normalized mean of exposed embeddings, not the gallery.

Cross-dataset key-reuse boundary

SCface: pool 3 rises from 5.29% to 33.17% top-1; fresh keys give 3.85%.

SCface pools 1/2/3 pass the all-seed interval criterion; 4/5/7/10 fail. Failure is not equivalence.

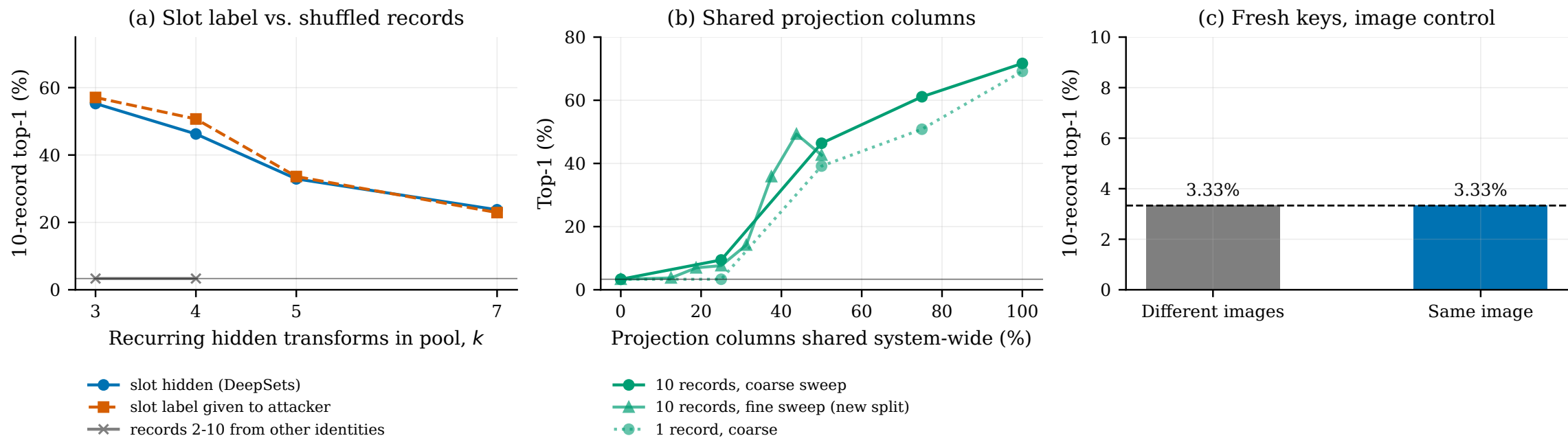


73 conditions / 12 source-separated, three-seed studies. Grey: unavailable. *: interval failure. Pilots excluded.

Mechanism controls and interpretation

MOBIO: slot labels change DeepSets by -0.83 to +4.44 points; shuffled records collapse to 3.33% chance.
Same-image fresh-key sets also give 3.33%. Projection-sharing effects depend on construction and partition.

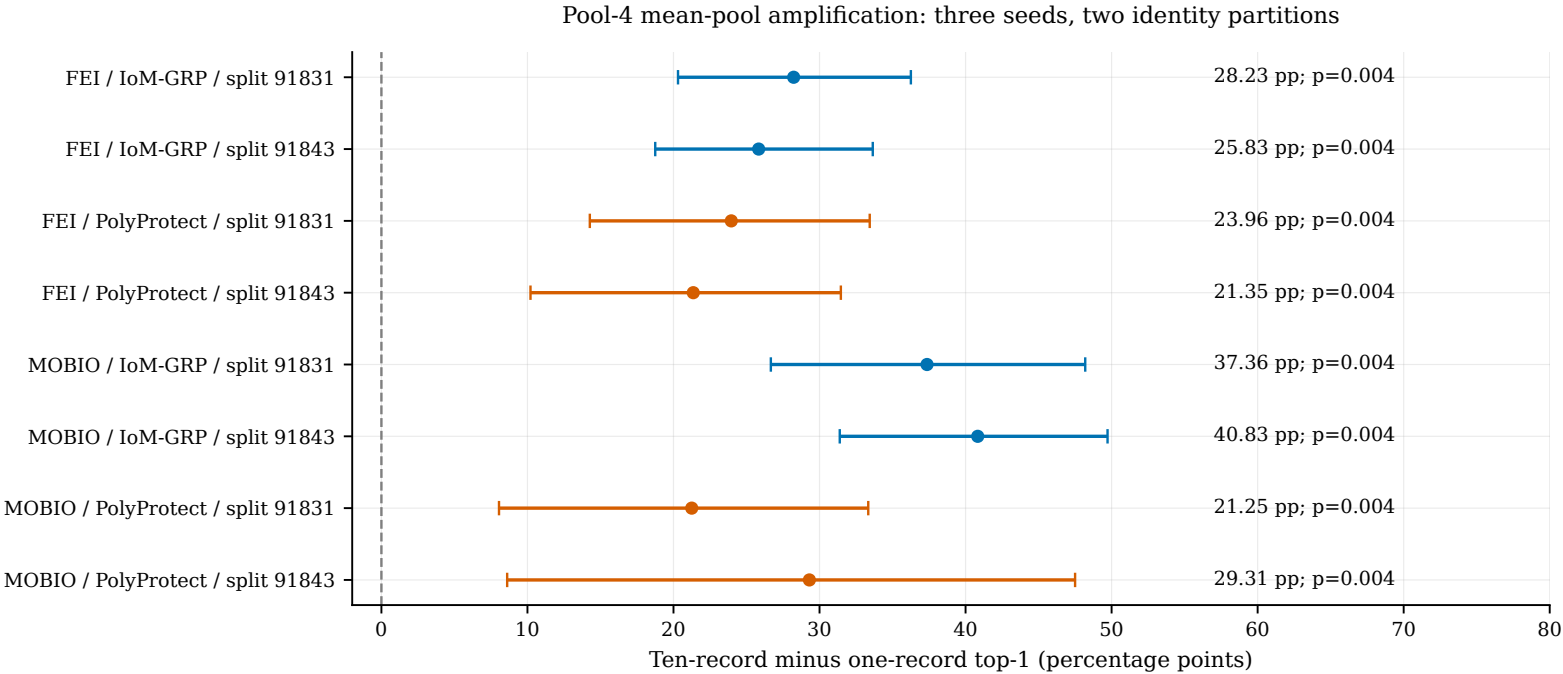
MOBIO mechanism controls (30 gallery identities; chance 3.33%)



Sources: experiments/mobio_mechanism_controls and mobio_correlation_controls. Three model seeds; chance 1/30.

Multi-seed, multi-partition validation

216 trained endpoints; fresh, shared and pool-4 keys; 3 model seeds x 2 identity partitions.
All 8 planned pool-4 amplification tests pass Holm correction (maximum adjusted $p = 0.004$).

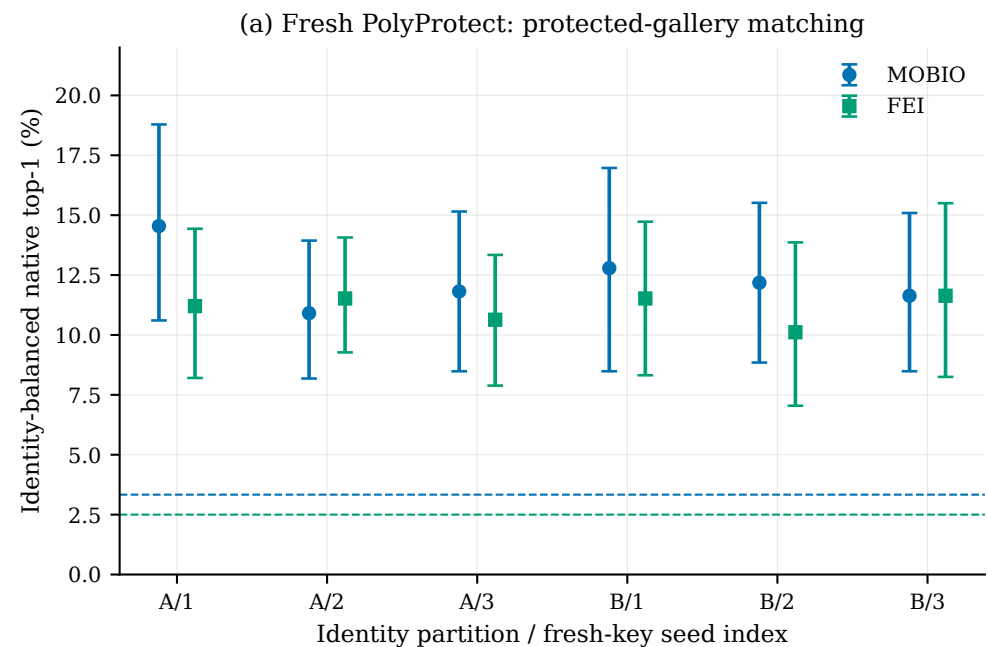


Bars: crossed model-seed / identity bootstrap 95% intervals. Tests: identity sign-flips, Holm family of 8.
Matched 120-epoch caps; key/set seeds fixed. Partitions are separate sensitivity analyses, not independent replications.

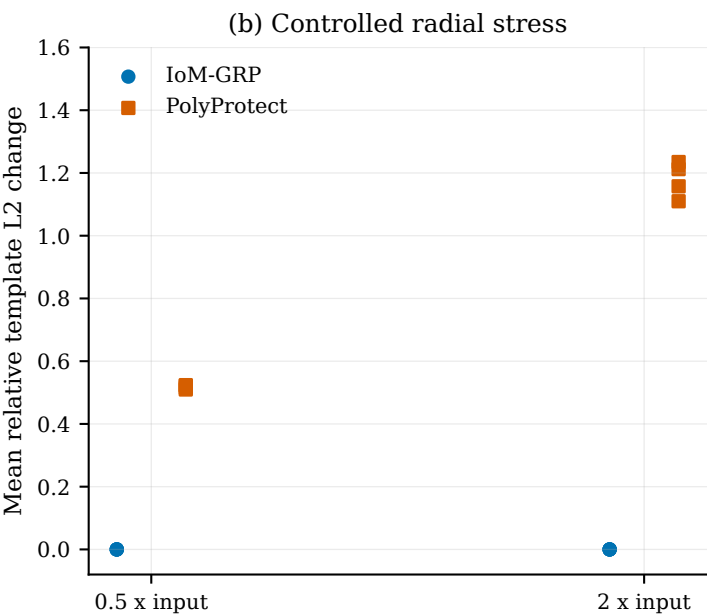
Matched 120-epoch caps. Crossed-bootstrap intervals include model-seed and identity resampling within each partition.

Native linkage and radial sensitivity

Fresh PolyProtect native identification exceeds the gallery-label null in 12 / 12 tests.
Maximum Holm-adjusted $p = 0.006$; three fresh-key seeds per dataset and identity partition.



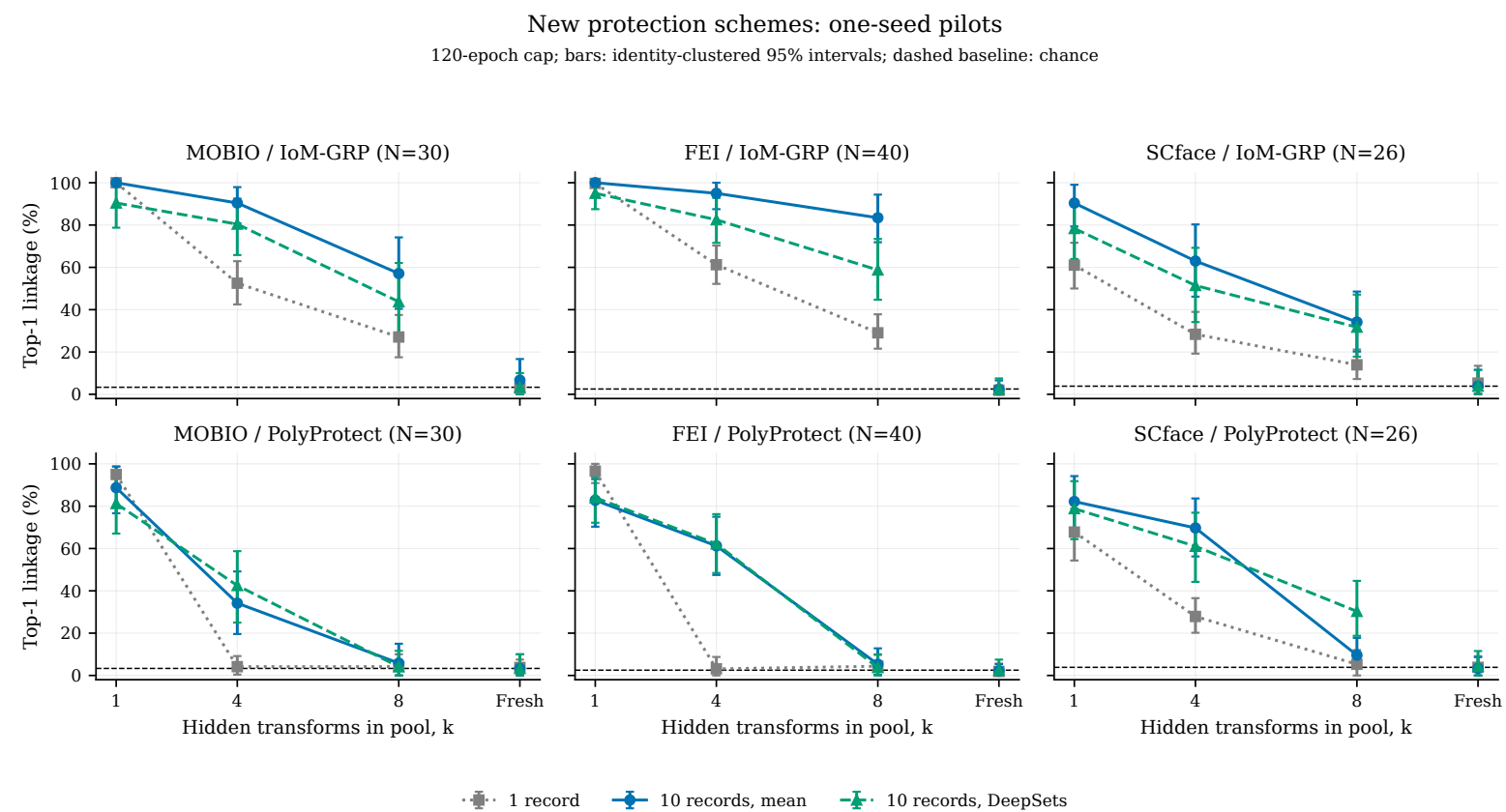
Native tests: 12 / 12 Holm-adjusted $p \leq 0.006$; identity-clustered 95% intervals; dashed lines: chance.
A/B: splits 91831/91843. Key seeds 1/2/3: 91873/91879/91883. Gallery-label permutations preserve probe clusters.
Radial stress: 32 held-out records per dataset/partition, fixed keys; measures scale sensitivity, not natural norm leakage.



IoM-GRP is invariant to the tested positive scales. PolyProtect is scale-sensitive; natural norm leakage was not tested.

Earlier cross-dataset scheme pilots

Earlier one-seed pilots: 72 model endpoints on MOBIO / FEI / SCface, including pool 8.
Separate native fresh-key PolyProtect diagnostic: 12.73% / 13.73% / 10.66% top-1, respectively.



Pilot intervals condition on one seed and partition; not confirmation. New replicated MOBIO/FEI results are on page 6.

Findings and study scope

TRANSFORM REUSE AND RECORD MULTIPLICITY

Recurring hidden transforms support substantial identity linkage from multiple protected records. The transition depends on dataset and identity partition, rather than a universal pool-size threshold. SCface pool 3: 5.29% single-record versus 33.17% ten-record top-1; chance is 3.846%.

MECHANISM

Shuffled-identity sets remove the MOBIO gain; record count alone does not explain the effect. Slot-label controls and partial projection sharing separate transform reuse from key disclosure. Mean pooling and DeepSets test aggregation at the template and learned-feature levels.

STATISTICAL AND THREAT-MODEL SCOPE

BioHash evidence spans four datasets and three model seeds per study. New scheme follow-up: 216 endpoints; all 8 primary contrasts pass Holm correction. Native protected-gallery matching and learned embedding linkage measure different attack surfaces. Fresh-key learned intervals include chance; they do not establish equivalence or general unlinkability.

Scope: fixed key/set seeds for training; two overlapping identity assignments; SCface remains one-seed for new schemes. Follow-up protocol and pre-execution hashes: `experiments/scheme_followup_2026-09-18/execution_manifest.json`.